

Course Guide

CS201 — Data Structures and Algorithms · هياكل البيانات والخوارزميات

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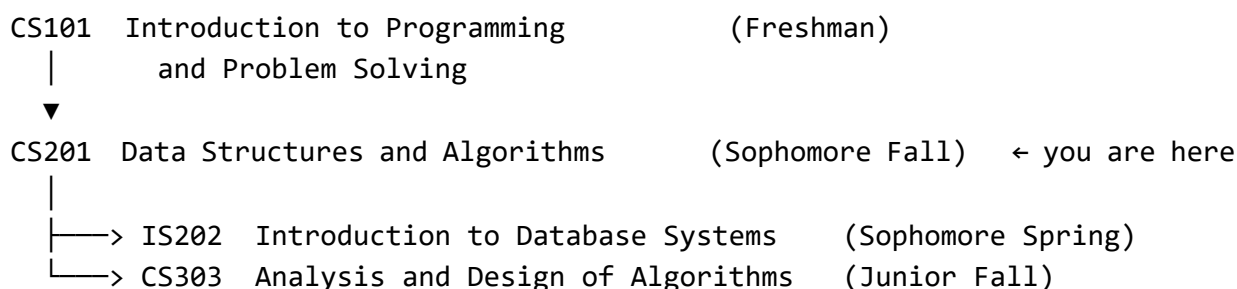
Every rule on this page is taken from the **FCIS Internal Bylaw for the Bachelor Stage, Credit Hour System, 2026**, with the page number printed next to it. The bylaw is in this repository: [regulations/FCIS-ByLaw-2026.pdf](#). Where this guide and the bylaw disagree, **the bylaw is right**.

1. The course

Field	Value	Source
Code	CS201	Bylaw p. 138
Name (EN)	Data Structures and Algorithms	p. 138
Name (AR)	هياكل البيانات والخوارزميات	p. 36
Credit hours	3	p. 138
Weekly contact hours	Lecture 2 · Tutorial — · Laboratory 2	p. 138
Prerequisite	Introduction to Programming and Problem Solving (CS101)	p. 138
Level / semester	المستوى الثاني (Sophomore) — Fall	p. 36
Required student workload	155 hours	p. 138
Equivalent ECTS	5.14	p. 138
Taught as a major in	IS, CS, IT, SE, MI, AI, NCS — all seven programs	p. 138
Taught as a minor in	—	p. 138

Official course content (quoted verbatim, bylaw p. 138)

"This course introduces fundamental data structures, algorithms, and abstract data types using a mix of programming and theory. Main topics include data structures such as arrays, lists, linked lists, stacks, queues, hash tables, heaps, priority queues, graphs, and trees. It introduces algorithms used for list manipulation, graph searches, sorting, searching, and tree traversals. Also, it introduces analyzing and managing the complexity associated with data structures and their operations."

Where CS201 sits

CS303 is where divide-and-conquer, greedy methods, **dynamic programming**, the master theorem, graph theory and NP-completeness are treated properly (bylaw p. 144). CS201 is the foundation it stands on.

2. How you are assessed

Taken directly from the CS201 specification, bylaw p. 138:

Component	Weight	Type
Student Activities / Practical Exam (SA/PE)	20	Exam
Midterm (MT)	15	Exam
Oral Exam (OE)	5	Exam
Final written exam (FE)	60	Exam
	100	

The two rules that fail students**1 — The double threshold** (Article 17, p. 23)

You must score **at least 60% of the total** and **at least 30% of the final exam mark**. Both. Missing either one is a fail.

*A student with 35/40 on coursework who scores 25% on the final has **failed**, even though the arithmetic total looks survivable. Coursework cannot rescue a final exam below 30%.*

2 — Attendance (Article 15, p. 21)

You must attend **at least 75%** of lectures and labs to be allowed into the final exam. Below that you are recorded as **محروم** (barred) and counted as a fail for the course.

Over a 14-week course that is roughly three sessions you can miss. Not five.

Grade scale (bylaw p. 22)

Grade	Points	Percentage	Grade	Points	Percentage
A+	4.0	$\geq 97\%$	C+	2.3	$73 - < 76\%$
A	4.0	$93 - < 97\%$	C	2.0	$70 - < 73\%$
A-	3.7	$89 - < 93\%$	C-	1.7	$67 - < 70\%$
B+	3.3	$84 - < 89\%$	D+	1.3	$64 - < 67\%$
B	3.0	$80 - < 84\%$	D	1.0	$60 - < 64\%$
B-	2.7	$76 - < 80\%$	F	0.0	< 60%

Non-GPA grades (p. 22): P ناجح · F راسب · W منسحب · Abs absent from the final without an accepted excuse · I incomplete.

The minimum passing grade in any course is **D** (p. 21). Graduation requires a cumulative GPA of at least **2.00 / 4.00** across **142 credit hours** (pp. 15–16).

3. Proposed — confirm before relying on this

The 20 marks for Student Activities / Practical Exam are allocated by the instructor, not by the bylaw. The split below is a proposal and is not final until announced in the lecture and on the WhatsApp channel.

Activity	Proposed marks
Lab exercises — pytest -m challenge passing, checked at milestones	10
Practical exam — implement a structure under time, in the lab	8
Homework 1 (environment + written comparison)	2
	20

Everything else on this page is bylaw text and is not negotiable.

4. What you need before week 2

```
git clone https://github.com/helghareeb/DSA27.git
cd DSA27

py -3.13 -m venv .venv
.\.venv\Scripts\activate
pip install -r requirements.txt

pytest -m "not challenge"          # every one of these must pass
```

You also need **Graphviz** installed as a program — the dot binary, not only the pip package. The environment tests check for it.

Full setup notes, including the Windows “python opens the Microsoft Store” trap and the VS Code debugging setup, are in the repository [README.md](#).

5. How the course runs

Lecture (2 h/week)	The idea, its cost, and why it exists.
Lab (2 h/week)	You implement it. The skeletons in <code>dsa/</code> raise <code>NotImplementedError</code> .
Tests	<code>tests/</code> defines what “correct” means. <code>pytest -m challenge</code> .
Visualisation	<code>viz/</code> draws, animates and measures. Already written — use it.

The rule that shapes everything, from `dsa/__init__.py`:

No module may use the built-in it is reimplementing.

`dsa/dynamic_array.py` may not be backed by a `list`. `dsa/hashmap.py` may not be backed by a `dict`. Routing around this may make tests pass; it will not make you a programmer.

Asking for help

1. Read the failing test. It states the contract.
2. Set a breakpoint and step through it — **F5**, *pytest: current test file*.
3. Then ask, on the [WhatsApp channel](#), with the failing test name and the error text.

“It does not work” is not a question. “`test_pop_from_empty_raises` expects `IndexError` and I am getting `None`” is.

6. Links

Repository	https://github.com/helghareeb/DSA27
WhatsApp channel	https://whatsapp.com/channel/0029Vb8XynEFy72KWbH6vS2V
YouTube	https://www.youtube.com/@0xHGH
Study plan	01-study-plan.md
Lecture 01	../lectures/01-why-this-course/lecture.md
Your program's regulations	regulations/
